

ECE 3070 Coursework (100%, including presentation [25%])

Tentative Submission Date: **23:59 26th May 2026 (Tuesday)**

Naming format: Zip all your files, Team#.zip, e-mail to xinboyu@intl.zju.edu.cn

In the report, show **team member info and contributions**.

I. Learning Objectives

- Introduce the varieties of medical image modalities.
- Understand the evaluation framework of medical image analysis.
- Gain knowledge and understanding of key research areas in machine learning for medical image analysis, including segmentation, detection, feature analysis and deep learning.
- Train practical skills in medical image analysis.
- Practice teamwork and leadership.
- Practice report writing and presentation skills.

The task is to carry out evaluations of methods in medical image analysis. You need to select and evaluate the performance of **three (or four)** models for a specific problem that you are interested in such as segmentation, classification, detection, etc.:

- The first one is from the classic image processing and **non-neural network** machine learning models, such as the active contour model, decision trees or random forests.
- The second one is from the **deep learning** models, such as the Mask RCNN, UNet, etc.
- The third one is an **open solution** that you learned from the Internet. (Teams with 3 members need to provide **2 open solutions**; accordingly, 2-member-team just need to provide one open solution.)

You are **NOT** recommended to choose the following datasets, which have been used in the last semester.

Brain MRI: <https://www.kaggle.com/datasets/mateuszbeda/lgg-mri-segmentation>

Cell Image: <https://www.kaggle.com/competitions/data-science-bowl-2018/data>

Retinal image: <https://drive.grand-challenge.org/>

We will check plagiarism based on last year's reports. You are recommended to choose any other public **medical related datasets** that you are interested in.

* You are allowed to use open-source code for the evaluations, but need to indicate the resources and how you contributed something new to the code's solution.

II. Submission

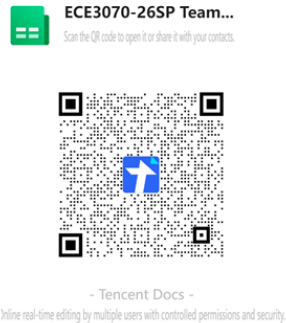
This is a group assignment

You will need to build a group with ideally **two** members (recommend different bachelor backgrounds) and a suggested workload allocation scheme is as follows:

1. Both two group members working on the code,
2. One member writing the word report,
3. One member preparing slides and doing the presentation.

Please **Login Tencent Docs** (腾讯文档) via the link or QR code as follows:

https://docs.qq.com/sheet/DVWRLS2xNdk5zbkN3?is_no_hook_redirect=1&tab=BB08J2



One of the member will need to submit the following items with **their mark weights**:

1. **Code (50%)**: Your exported Jupyter notebook submitted as an .ipynb file.
 2. **Report (25%)**
 3. **Slides & Presentation (25%)**
- * Export all files as PDF and package them as a zipped file, then submit it through a Backboard link.

III. Evaluation Criteria

Code Evaluation Criteria (100 marks):

Jupyter Notebook:

1. Data sample visualization, (20%)
2. Model explanation: reference, package, comments, etc. (20%)
3. Training, validation, You need to cover both the training (where appropriate) and evaluation, record the model performance adequately with relevant metric. (30%)
4. Results, pay attention to the detailed progress regarding how you get it. And highlight the results such as tables or visualized figures that you incorporated in the report. (30%)

Report Evaluation Criteria (100 marks):

The main body of each report should be around 2000-2500 words in addition to text (which is about 5-6 pages). The main report should be structured as follows:

A Cover Page (5%)

- Team Information: student numbers, names, etc.
- **Each member's contribution.** (will adjust the marks based on this if necessary)

Overview/Summary (15%)

- The task
- Your name and student numbers and class number
- Briefly describe what you did
- Provide a summary of your main findings

Method (20%)

- Describe the data processing, feature extraction, etc. performed (and why it was performed)
- Describe the models used (and why they were selected)
- Describe any other things that you did or tried in order to improve performance

Training and Validating etc. (20%)

- Describe the training schedule and approach that you undertook.
- Show your working here – where you report all your training and validation, etc. that you performed in order to get the results.
- Note that it is important that your results can be replicated. All code to reproduce the final predictions must be included, along with any code that justifies your choices.

Results and Discussion (20%)

- Present your findings for each of the approaches tried.
- Create a table that lists the approaches and the training and testing performances (along with relevant parameter settings).
- Explain your findings, what worked, what didn't, and why? Which features were the most important? When does the model fail?
- Add in any additional analysis etc that you performed here.

Summary and Recommendation (15%)

- Provide a recommendation of which approach(s) should be used/considered and the pro's and con's of the approach(s).
- The future works based on your best understanding from perspectives such as data, model, and optimization perspectives.

References (5%)

- List relevant references and resources

Presentation Criteria (100 marks):

Presentation score = standard rubric score + grading quality

Standard Rubric (80%), calculation is:

$$\text{standard rubric score} = (\text{peer average} + 2 \text{ lecturers' grading})/3$$

Revised from: <https://courses.engr.illinois.edu/cs598hs/sp2020/rubrics/rubric-presentation.pdf>

Components	Weight				Components	Weight			
Organization	10%	Presentation is clear, logical, and organized. Listener can follow line of reasoning.	Presentation is generally clear and well organized. A few minor points may be confusing.	Organization is highlighted. Listener can follow presentation only with effort. Arguments are not clear.	Design	10%	Communication aids enhance presentation. • The text on the visuals is readable. • Information is represented and organized to maximize audience comprehension. • Details are highlighted so that main points stand out.	Communication aids contribute to the quality of the presentation. • Font size is mostly readable. • Appropriate information is included. • Some material is not supported by visual aids.	Communication aids are poorly prepared or used inappropriately. • Font size is too small to read. • Too much information is included. • Details or some unimportant information is highlighted, and may confuse the audience.
Content	30%	Speaker provides accurate and complete explanations of key concepts and theories. • An overview of related work is provided; connections to the paper being presented are clear. The complete reference for every paper is shown on screen. • Applications of theory illuminate issues. • Equations are carefully presented, with all variables and constants explained on the slide. • The key insight behind the technical innovation is communicated with examples, or analogies. • Strengths and weaknesses are highlighted; new ideas that build upon strengths and mechanisms to overcome weaknesses are discussed. • Listeners gain insights.	For the most part, explanations of concepts and theories are accurate and complete. • Some related work is presented, but relationship to the paper being presented is unclear. Only a partial reference is shown on the slide. • Some logical applications are included. • Equations are presented to illuminate the main idea, but the meaning of the variables and constants is unclear. • The technical innovation is presented but there is little attempt to be insightful; examples, analogies are absent. • Brief mention of strengths and weaknesses, but little attempt to amplify strengths or to overcome weaknesses.	Explanations of concepts and/or theories are inaccurate or incomplete. • Related work is either absent or briefly referenced without explanation; The reference is missing from the slide. • Equations are either absent, or when presented are glossed over. • Little attempt is made to present the main innovation. • Strengths and weaknesses are not discussed. • Little attempt is made to tie the theory to applications. • Listeners gain little from the presentation.	Grammar & Language	10%	Sentences are complete and grammatical. They flow together easily. Words are well chosen; they express the intended meaning precisely.	Sentences are complete and grammatical for the most part. They flow together easily. With some exceptions, words are well chosen and precise.	Listeners can follow presentation, but they are distracted by some grammatical errors and use of slang. Some sentences are halting, incomplete, or vocabulary is limited or inappropriate.
					Audience Interaction	10%	Consistently clarifies, rectifies, and responds to questions. Summarizes when needed.	Generally responsive to audience questions and needs. Misses some opportunities for interaction.	Responds to questions inadequately.
Style	10%	Level of presentation is appropriate for the audience. Presentation is a planned conversation, geared for audience understanding. It is not a reading of a paper. Speaker is comfortable in front of the group and can be heard by all.	Level of presentation is generally appropriate. Pacing is sometimes too fast or too slow. Presenter seems slightly uncomfortable at times, and audience occasionally has trouble hearing him/her.	Aspects of presentation are too elementary or too sophisticated for audience. Presenter seems uncomfortable and can be heard only if listener is very attentive. Much of the information is read.	Involvement	20%	Based on your grading quality to all the other teams.		

Adapted from the "Oral Communication Rubric", Elderly Center for Teaching Excellence, Carnegie Mellon University

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Peer assessment involvement (20 %):

Inter-group: The quality of the inter-group assessment is based on the bias to the relatively unbiased grading, taking the average grading quality for all the other teams, as follows:

$$grading\ quality = \frac{1}{7} \sum_{t=1}^7 20 - \frac{20}{100} \times |your\ grading_t - standard\ rubric\ score_t|$$

Remark: If your team has conflict and cannot continue to work together, each member will be deducted some marks based on the situation and lecturers' investigation.

Your grade also depends on your teamwork. If your team cannot work well together, the instructors will initiate an investigation, which will determine your final grade. This is likely to have a negative impact on both members due to the lack of collaborative work.